MILITARY SPECIFICATION

ACCESSORY KIT; WARNING DEVICE, DEMOUNTABLE, FOR EMERGENCY VEHICLES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 14 volt and 28 volt demountable warning device consisting of a support bar and brackets for rain gutter mounting and gutterless mounting of two rotating warning lights, speaker, electrical harness, electronic siren and public address (P/A) system with microphone.

1.2 Classification.

1.2.1 Types. Warning kits shall be of the following types, as specified:

| | |
|---|---|
| Type I | 14 Volt DC |
| Type II | 28 Volt DC |

1.2.2 Warning lamp domes. The kits shall be supplied with interchangeable warning light domes of the following colors as listed in MIL-L-45325:

| | |
|---|---|
| Class 1 | Amber Dome |
| Class 2 | Blue Dome |
| Class 3 | Red Dome |

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-585 | - Boxes, Wood, Wirebound |
| PPP-B-591 | - Boxes, Fiberboard, Wood - Cleated |
| PPP-B-601 | - Boxes, Wood, Cleated - Plywood |
| PPP-B-621 | - Boxes, Wood Nailed and Lock - Corner |
| PPP-B-636 | - Box, Fiberboard |



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Military

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of. |
| MIL-C-34320 | - Cable |
| MIL-L-45325 | - Light, Warning, Vehicular: Rotating Unit, 14 and 28 Volt DC |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedure and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-202 | - Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment. |
| MIL-STD-462 | - Electromagnetic Interference Characteristics Measurement of. |
| MS-51317 | - Light Warning, Vehicular, Rotating, Direct Current |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS

Publication

| | |
|---|---|
| SAE J558 | - Low Tension Cable |

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 485 Lexington Ave , New York, N. Y. 10017.)



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AMERICAN STANDARDS ASSOCIATION

PUBLICATION

| | |
|---|---|
| S1.2 | - Physical Measurement of Sound, Method of Octave-Band Filter Set for. |
| Z24.10 | - Analysis of Noise and Other Sounds, Specification for. |
| S1.4 | - General - purpose Sound Level Meters, Specification for. |

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, NY 10016.)

3. REQUIREMENTS

3.1 Preproduction sample. Preproduction samples of the warning kits to be furnished under this specification shall be produced prior to the manufacture of these items in production quantity. The manufacturer shall subject these warning kits to the preproduction inspection specified herein to determine conformance to the quality assurance provisions of this specification (see 4.2). The preproduction samples shall be fully representative of the items to be supplied from production facilities and tooling.

3.2 Materials. Materials shall be of such quality as to be free from defects that might adversely affect performance. Materials having a tendency to warp or to absorb moisture shall not be used.

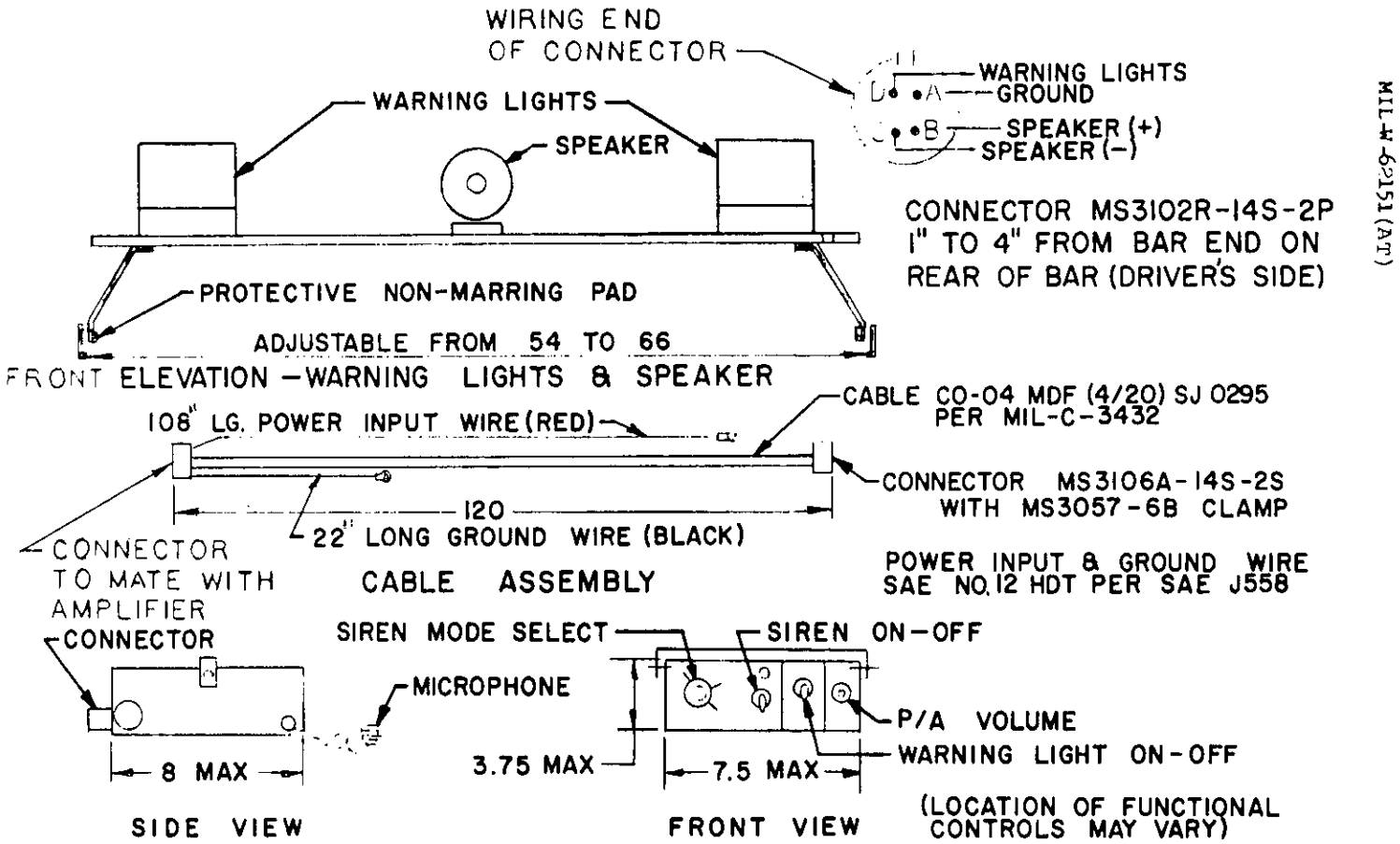
3.2 1 Painted finish. When specified, a painted finish shall be applied in accordance with detail requirements furnished by the procuring activity.

3.2.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.3 Design and construction. The demountable warning device shall conform to Fig. 1. Soldering shall conform to requirement 5 of MIL-STD-454



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ELECTRONIC SIREN & AMPLIFIER

NOTE: ALL DIMENSIONS IN INCHES

ACCESSORY KIT; WARNING DEVICE FOR EMERGENCY VEHICLES

FIGURE 1



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3.3.1 Rotating warning lights. The rotating warning lights shall meet the requirements of MIL-C-45325 and MS-51317 except that the maximum skirt diameter shall be 8.75 inches or less and the maximum height shall be 8.25 inches or less. The lamps shall have a plate completely enclosing the bottom of the housing for mounting onto the bar.

3.3.2 Speaker. A round (as contrasted to retangular) speaker, having an impedance of between 13 and 16 ohms, shall be supplied with a driver rating of 75 watts (nominal). The speaker shall be weatherproof and corrosion resistant with a mounting plate provided for attachment to the bar.

3.3.3 Mounting bar. The mounting bar must be mountable on gutterless vehicles or vehicles with conventional rain gutters and it shall be adjustable to accommodate gutters spaced (or roof widths) from 54 to 66 inches. All wiring for the warning lights and speaker shall be inside the bar or component with adequate allowances for the full range of bar adjustment. Attaching hardware shall be weather and corrosion resistant, and wire holes in the bar shall be fitted with grommets. Mounting clamps must be tamperproof as far as practible (employ some method of attachment that discourages vandalism).

3.3.4 Electronic siren and public address system. Separate on-off switches shall be provided for the siren and warning lights. With the siren switch "ON" either the Alert (yelp), Wail (coaster), HI-LO of P/A (public address system) must be selectable. The microphone shall be of the noise cancelling type (magnetic) to limit feed back. The microphone cord shall be retractable and eight feet in length. All control knobs shall be functionally marked. Instructions shall be clearly and permanently displayed on the control unit. All knobs and functional markings on the unit shall be illuminated for night viewing with adequate contrast and brightness. The sound intensity of the system shall be a minimum of 115 decibels at six feet from the speaker on the speaker axis. Warning light and siren circuits shall be fused separately. In the public address mode a zero to full attenuation volume control shall be provided.

3 3.4.1 Siren tone range.

The ratio of the highest fundamental frequency to the lowest shall be 3 to 1. The highest fundamental frequency shall not exceed 1500 Hz.



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3.3.4.2 Siren tone modulation

a. Wail - The tone range shall be accomplished in a steady rise and fall in 5 to 7 seconds and repeated indefinitely.

b. Yelp - The tone range shall be accomplished in a steady rise and fall in 0.3 to 0.1 seconds and repeated indefinitely.

3.3.4.3 Public address system

Audio frequency response of the public address system shall be flat within ± 3db from 250 to 4000 Hz at rated power.

3.3.5 Harness. Cable for the wire harness (mounting bar to electronic siren) shall be type CO-04 MOF (4/20) SJ 0295 per MIL-C-34320 (see Fig. 1). The cable connector (bar connection) shall be type MS-3106A-14S-2S with clamp MS-3057B and the power input and ground wire shall conform to SAE No. 12 HDT per SAE-J-558.

3.4 Instructions.

3.4.1 Installation instructions. Each unit shall have included complete installation instructions covering assembly, mounting and interconnection.

3.4.2 Operating instructions. Each kit shall be furnished with complete operating instructions or manual.

3.4.3 Maintenance instructions. There shall be included with each unit complete maintenance instructions, electrical schematic diagram and parts list for all components.

3.5 Performance.

3.5.1 Warning lights. The warning lights shall meet the performance requirements of MIL-C-45325.

3.5.2 Electronic siren/public address system. When tested as specified in 4.6.2, the sound intensity of the system shall be a minimum of 115 decibels at a distance of six feet from the speaker on the speaker axis.



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3.6 Environmental.

3.6.1 Extreme temperature. The system shall provide continuous standby operation at ambient temperatures from - 50° to 140°F. A maximum warm-up time of 60 seconds is permitted.

3.6.2 Vibration. After testing as specified in 4.7.2, the system shall operate without malfunction and shall evidence no permanently deformed, cracked, or broken parts.

3.6.3 Corrosion. When tested as specified in 4.7.3 the system (excluding the control unit) shall show no corrosion which affects appearance or performance of the system.

3.6.4 Electromagnetic interference. When tested as specified in 4.7.4 the system shall meet the requirements of CE07 and RE05 of MIL-L-STD-461 for tatical and special purpose vehicles and engine driven equipment.

3.7 Marking. Marking shall conform to MIL-STD-130 and as a minimum shall include military part number, voltage rating and manufacturer's identification.

3.8 Workmanship. Workmanship shall be such as to assure a product free of any malfunction, loose hardware, rust, scratches, burrs, sharp edges, chips or foreign materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.



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4.2.1 Warning kit samples. Preproduction samples shall consist of two complete warning kits. The samples shall be representative of the items proposed to be furnished to the Government. Preproduction inspection shall be performed by the contractor under Government surveillance and consist of examination for defects specified in Table II and testing as specified in Table I.

Table I Order of preproduction testing

| Specimen | Paragraph | Test |
|---|---|---|
| 1 | 4.6.1 | Operation |
| | 4.6.2 | Sound intensity |
| | 4.7.1.1 | Low temperature |
| | 4.7.1.2 | High temperature |
| | 4.7.1.3 | Vibration |
| | 4.6.1 | Operation |
| 2 | 4.6.1 | Operation |
| | 4.6.2 | Sound intensity |
| | 4.7.4 | Electromagnetic interference |
| | 4.7.3 | Salt fog (corrosion) |
| | 4.6.1 | Operation |

4.2.2 Failure. Failure of a preproduction sample to pass any of the examinations or tests specified herein may be cause for the Government's refusal to witness additional examinations or tests until the faults revealed have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all warning kits of one type, class and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.



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4.3.1.3 Sampling for testing. A representative sample shall be selected in accordance with level S-3 of MIL-STD-105 from each lot that has passed the examination specified in 4.3.2.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

| Classification | AQL |
|---|---|
| Major | 1.0 |
| Minor | 2.5 |

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in Table II.

Table II Classification of defects

| Categories | Defects |
|---|---|
| Major | |
| 101 | Improper or defective materials used (see 3.2) |
| 102 | Warning lights do not conform to MS-51317 (see 3.3.1) |
| 103 | Accessory kit not properly marked (see 3 7) |
| Minor | |
| 201 | Poor workmanship (see 3.8) |

4.3.3 Testing.

4.3.3.1 Quality Conformance test. Samples selected in accordance with 4.3 1.3 shall be subjected to the test specified in 4.6.1 using an AQL of 2.5 on the basis of percent defective.

4.4 Control test.



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4.4.1 Sampling for control test. The warning kits shall be selected at the rate of one of each 500 produced or fraction thereof, providing that no more than three units shall be selected in any 30-day period. The control test sample shall be identified as to production period and after having been examined for the defects specified in 4.3 2.2, they shall be subjected to the tests specified in 4.6.1, 4.6.2, 4.7.1.1, 4.7.1.2, and 4.7.3.

4.4 2 Failure. Failure of any control test sample to pass any examination or test specified herein may be cause, at the option of the Government, for refusal to accept subsequent lots until the faults revealed by the test have been corrected.

4.5 Test procedures.

4.5.1 General.

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, tests shall be conducted at ambient room temperature of $77^{\circ} \pm 15^{\circ}$F.

4.5.1.2 Calibration of test equipment. Test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance. Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government.

4.6 Performance tests.

4.6.1 Operation. The system shall be assembled and connected to a 14 $\pm$ 0.5 volt or 28 $\pm$ 0.5 volt DC power source as applicable. All the functions of the system shall be checked for operation including warning lights, siren yelps, siren wail, siren Hi-Lo and public address. The mounting bar shall be extended to maximum mounting width and contracted to minimum mounting width. All functions shall operate satisfactorily without interruption.



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4.6.2 Sound intensity. The sound intensity of the system shall be determined and shall be, as a minimum, 110 decibels when measured at a distance of six feet from the speaker on the speaker axis. This test should be conducted in an anechoic chamber. All siren system modes shall be tested. Test equipment, procedures, and standards shall be in accordance with American Standards Association documents: S1.2, Z24.10 and S1.4.

4.7 Enviromental tests.

4.7.1. Effects of extreme temperature.

4.7.1.1 Low temperature. The system shall be placed in a test chamber and electrical circuits connected. The temperature of the chamber shall be lowered to - $50^{\circ} \pm 5^{\circ}$F and the system shall soak at that temperature for a period of 6 hours. After six-hour period the functions of the system shall be energized and operation shall commence within 60 seconds. Also, while at this temperature components and fittings shall be checked for fit.

4.7.1.2 High temperature. The system shall be placed in a test chamber and electrical circuits connected. The temperature of the chamber shall be set at $140^{\circ} \pm 5^{\circ}$F and the system shall soak at that temperature for a period of 6 hours. After the six hour period the function of the system shall be energized and operation shall commence within 60 seconds. The domes of the warning lights should be removed for this test.

4.7.2 Vibration. The system excluding the control unit shall be subjected to the vibration test in accordance with method 201 of MIL-STD-202 except the frequency shall be 10 to 30 hertz (Hz) and the time of vibration in each of three mutually perpendicular directions shall be one hour (total of three hours).

4.7.3 Corrosion. The speaker and mounting bar shall be subjected to salt fog test method 101, condition B, of MIL-STD-202. After testing the item shall be inspected for corrosion and operation.

4.7.4 Electromagnetic interference. To determine conformance to 3.6.4, the item shall be tested in accordance with MIL-STD-462.

4.8 Inspection of preparation for delivery.



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4.8 Inspection of preparation for delivery.

4.8.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to the requirements of section 5. Any evidence of deviation from specified requirements shall be cause, at the option of the Government, for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each warning kit shall be preserved in accordance with MIL-P-116 method III, and individually packaged in containers conforming to PPP-B-636, class weather resistant. Closure, sealing and taping, for waterproofing, shall be in accordance with the appendix to the box specification.

5.1.2 Level C. This is the degree required for protection under known favorable conditions during a shipment, handling and limited tenure of storage. Preservation-packaging and packing designated level C shall be designed to protect items against physical and environmental damage during known favorable condition of shipment, handling and storage. In general, the following will determine the requirements of level C.

(1) Limited handling during transportation and intransit storage.
(2) Limited shock, vibration and static loading during transportation cycle.
(3) Controlled warehouse environment for temporary periods.
(4) Effect of environmental exposure during shipment and intransit delays.
(5) Stacking and supporting superimposed loads during shipment and temporary storage.
(6) Items characteristics require no special or peculiar preservation-packaging and/or packing provisions.



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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Unit packages shall be packed in a wirebound wood box, fiberboard wood cleated box, wood cleated plywood box or nailed wood box, conforming to PPP-B-585, style 2 or 3, class 3, PPP-B-591, style A or B, class II, PPP-B-601, style A or B, overseas type, or PPP-B-621, style 2, 2 1/2, class 2, respectively. Box shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Gross weight and size shall not exceed the limitations of the container utilized.

5.2.2 Level B. Warning kits preserved and packaged as specified in 5.1.1 shall be packed in boxes conforming to PPP-B-636, CF or SF class domestic. Closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. This is the degree required for protection under known favorable conditions during shipment, handling and limited tenure of storage. Preservation-packaging and packing designated level C shall be designed to protect items against physical and environmental damage during favorable conditions of shipment, handling and storage. In general, the following criteria will determine the requirement of level C.

(1) Limited handling during transportation and intransit storage.
(2) Limited shock, vibration and static loading during the transportation cycle.
(3) Controlled warehouse environment for temporary periods.
(4) Effects of environmental exposure during shipment and intransit delays.
(5) Stacking and supporting superimposed loads during shipment and temporary storage.
(6) Item characteristics require no special or peculiar preservation-packaging and/or packing provisions.



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5.3 Marking. In addition to any special marking required by the contract or order, initial carton, unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The demountable warning device is intended for use on security vehicles and other emergency vehicles such as commercial sedans, vans or pickup trucks (with or without rain gutters) having 12 or 24 volt electrical systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Painted finish if required (see 3.2.1).
(d) Applicable Military standard (see 3.3.1).
(e) Whether or not permanent markings are required (see 3.7).
(f) Level of packaging (see 5.1) and packing (see 5.2) required.

Custodians:
Army - AT

Review Activity:
Army - ME

Preparing Activity
Army - AT

Project No. 6220-A222



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-W-62151(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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